

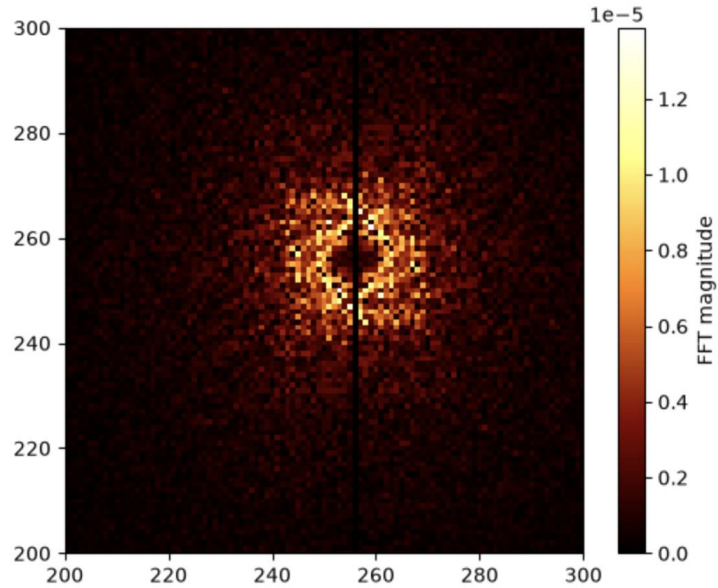
Week 2 Progress Meeting presentation

By Poline S

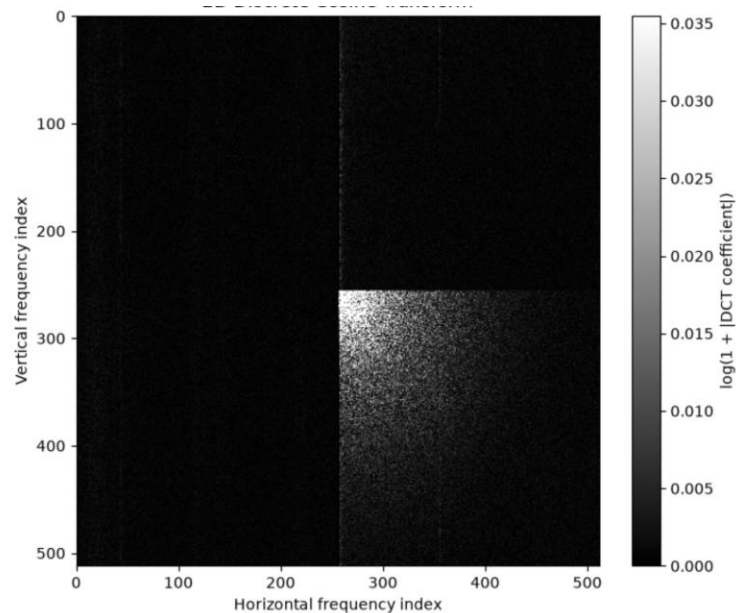
Main changes to the code

- New app added to display dots and when clicking on a dot, the app opens a window with dot parameters (Using the LoG method of dot detection). Now working on allowing the user to select the algorithm used for dot detection
- Looked into more space transformation:
 - DST, Discrete Harley, Gabour wavelets (surprisingly, this worked pretty well! And curious to see the relationship between Gabour wavelet response and actual dot probability)
- Calculated Otsu's criterion for LoG and others (lower bound is 0.375 and we 'eyeballed' it to be 0.4 which may be why the dot algorithm was detecting dots so well last time)
- Looked into different types of smoothing
 - Gaussian smoothing (this worked last time), added anisotropic diffusion, mathematical morphology has also worked well last time with erosion and dilation within the top hat applied filter
- For low level feature extraction, last time we used LoG (Mexican Hat and the Blob function)
 - Added phase congruency and SIFT for dot detection and saliency for searching for any other interesting things on the plot
- Added calculations for dot circularity via shape matching.

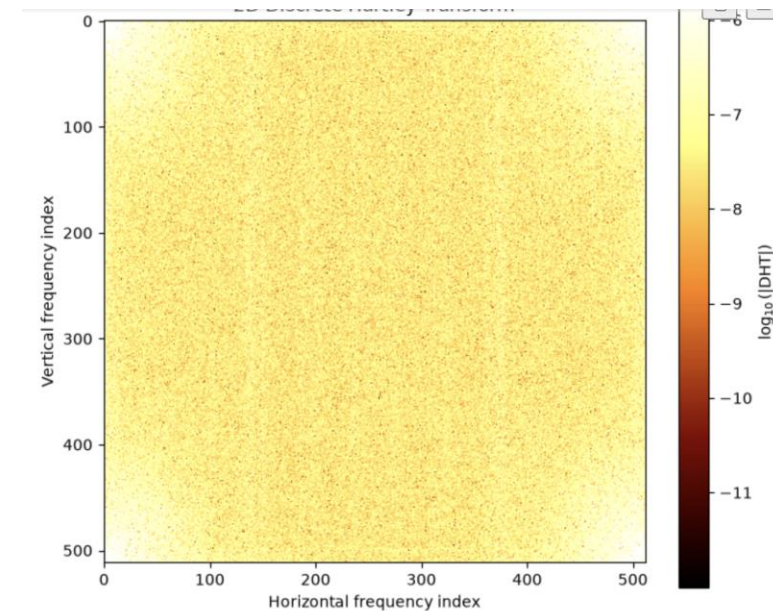
1.1 Looking into different transforms



Fourier Transform (seen last time)



DCT

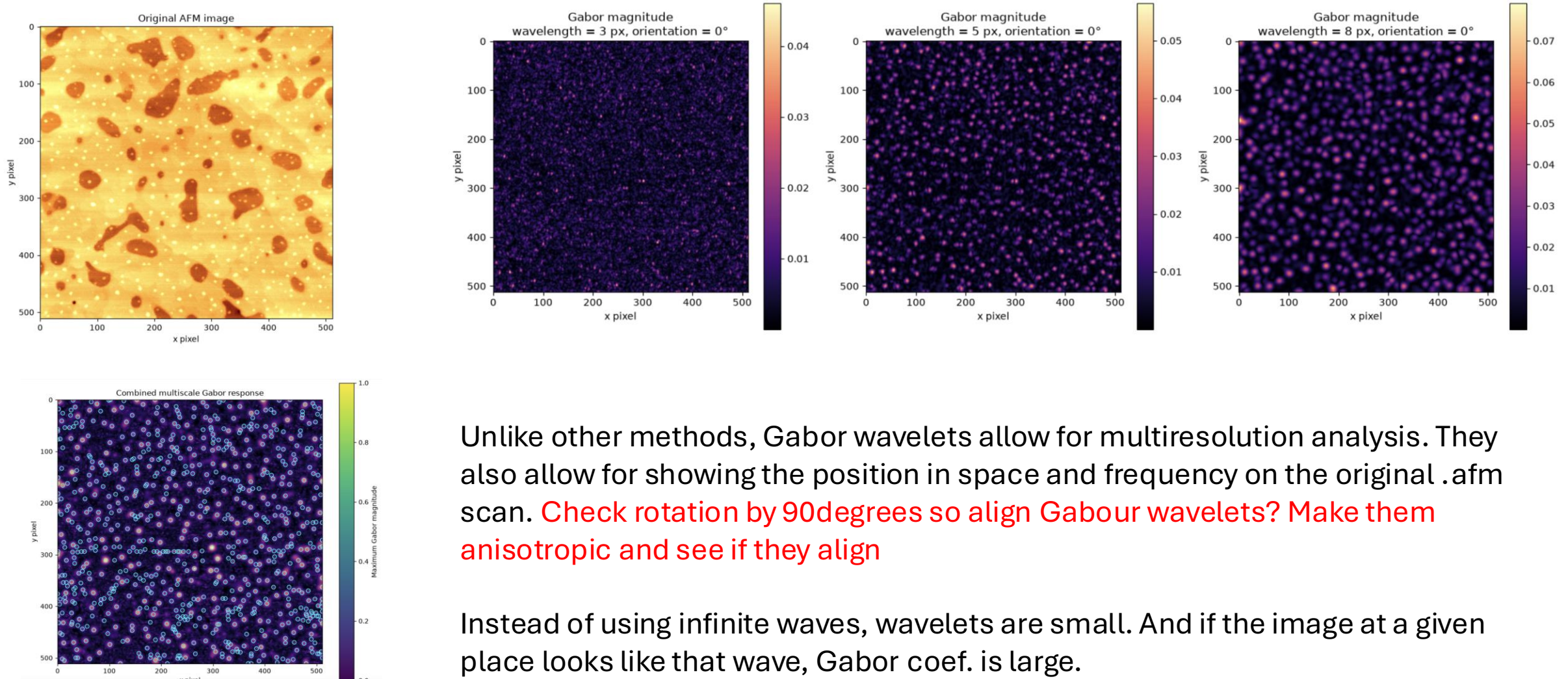


Hartley

All very similar to each other, but DCT transforms space into less coordinates. So, **when training the model we may need to use DCT instead of FT to minimise the memory used.** Hartley simplifies complex number calculations but this came out not to matter as much.

1.2 Gabor Wavelets for QD detection

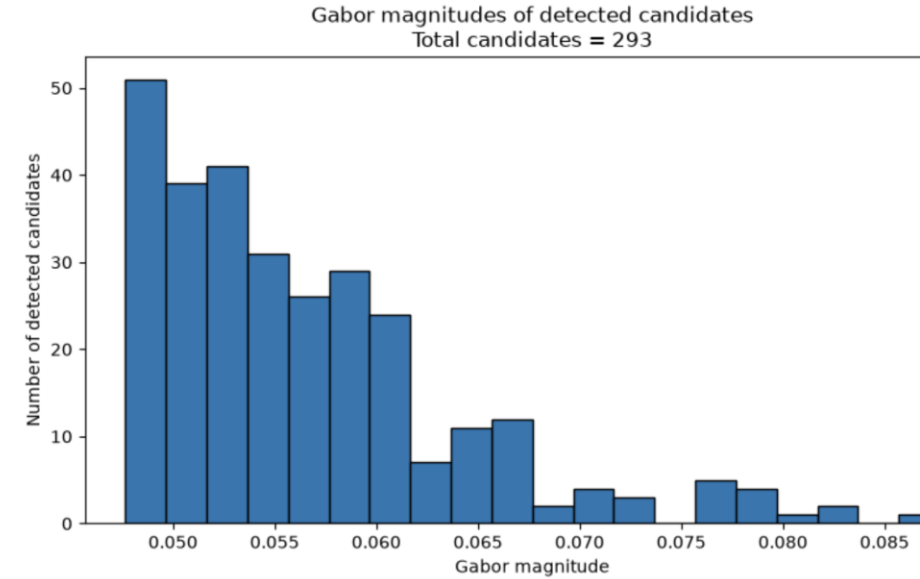
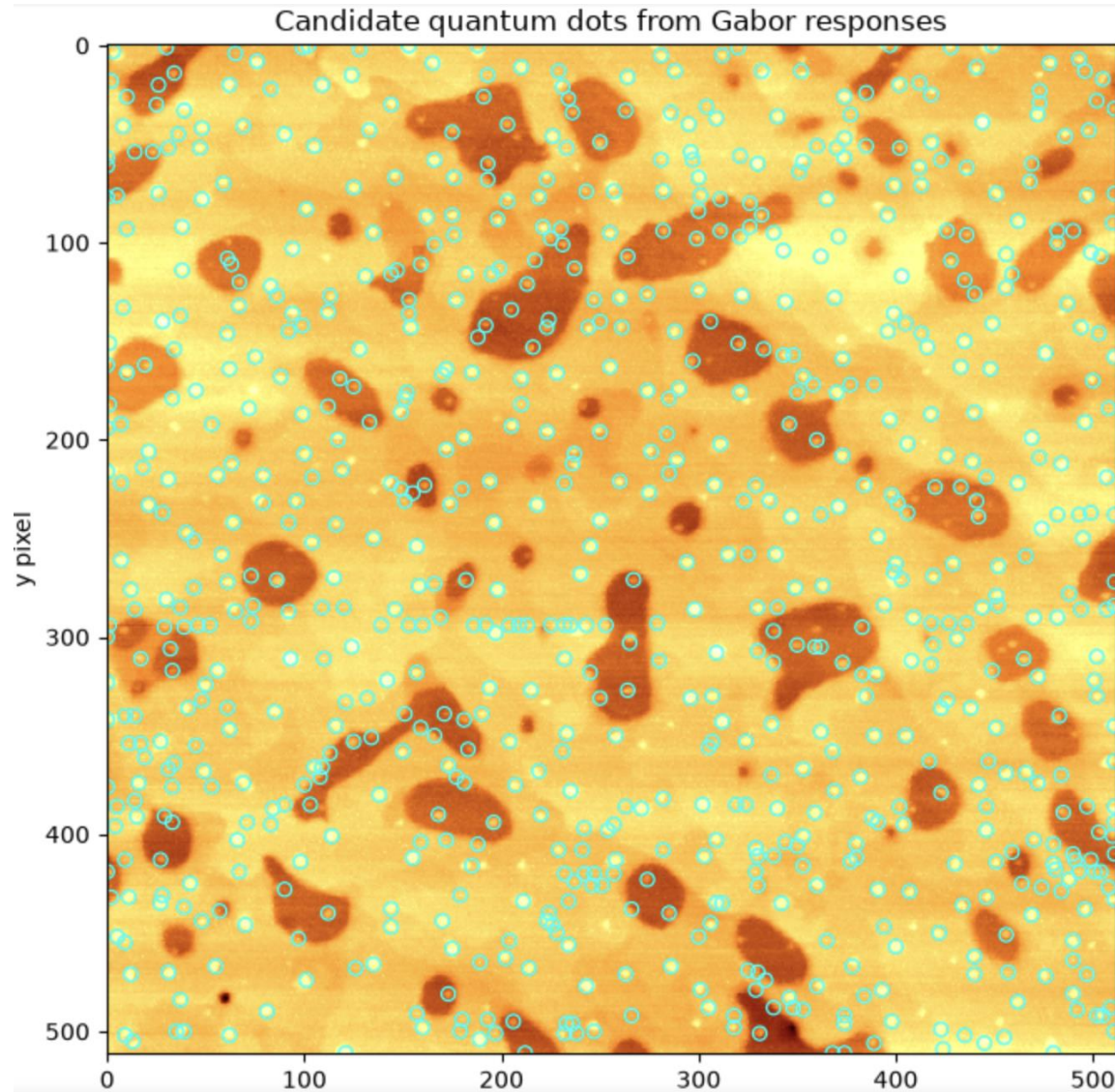
Try image
equalisation
through its
hystogram



Unlike other methods, Gabor wavelets allow for multiresolution analysis. They also allow for showing the position in space and frequency on the original .afm scan. Check rotation by 90degrees so align Gabour wavelets? Make them anisotropic and see if they align

Instead of using infinite waves, wavelets are small. And if the image at a given place looks like that wave, Gabor coef. is large.

1.3 Comparing Gabor wavelets probabilities



Make
Gabor
work less
with
horizontal
defects ->
may be a
artefact
streak.
Detected
lattening

$$gw(t, w_0, t_0, a) = e^{-j\omega_0 t} e^{-\left(\frac{t-t_0}{a}\right)^2}$$

position
 modulating frequency
 letting $f_0 \rightarrow 0$ removes frequency information.
 width of the gaussian envelope
 Can return to FT by setting a very large

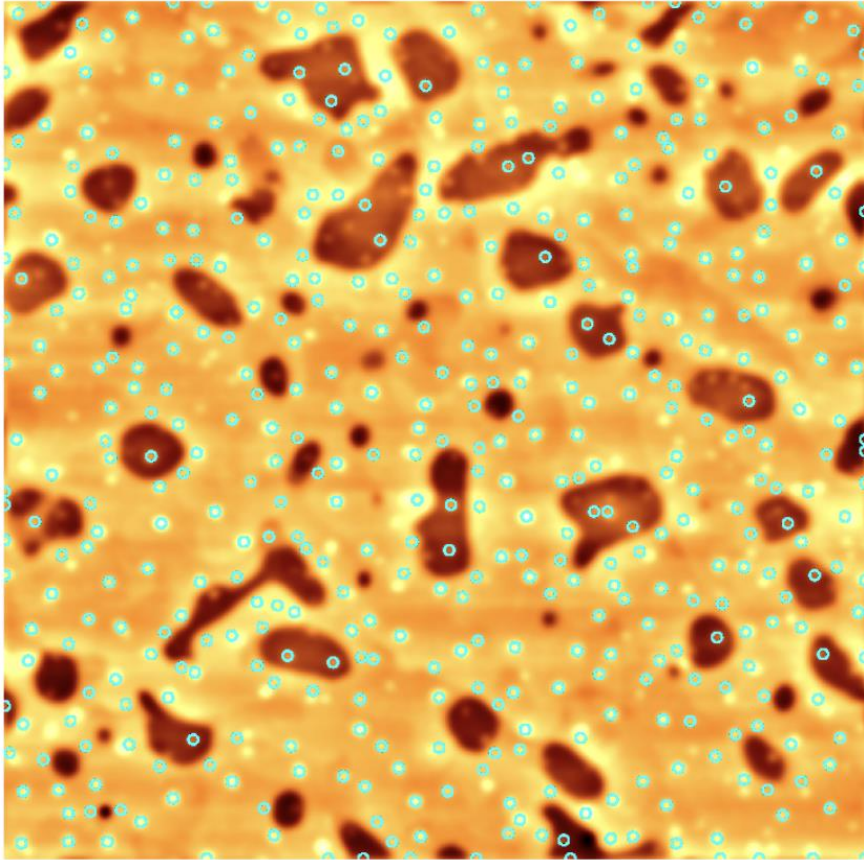
$$gw_{2D}(x, y) = \frac{1}{\sigma\sqrt{\pi}} e^{-\left(\frac{(x-x_0)^2 + (y-y_0)^2}{2\sigma^2}\right)} e^{-j2\pi f_0 ((x-x_0)\cos\theta + (y-y_0)\sin\theta)}$$

controls the direction (orientation) of the wavelet

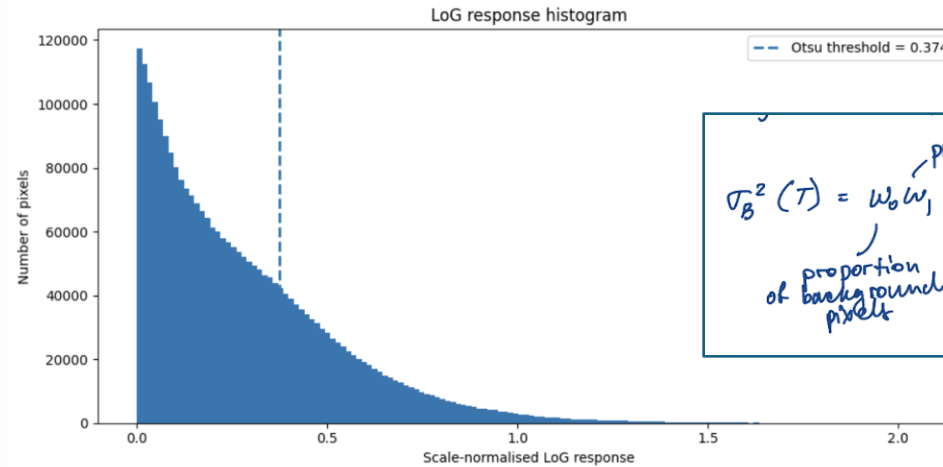
Better
flattening
methods.

2.1 Thresholding – Otsu lower cutoff

Otsu threshold = 0.374
Detected blobs = 489



Otsu threshold: 0.3742742285724291
Detected blobs: 489



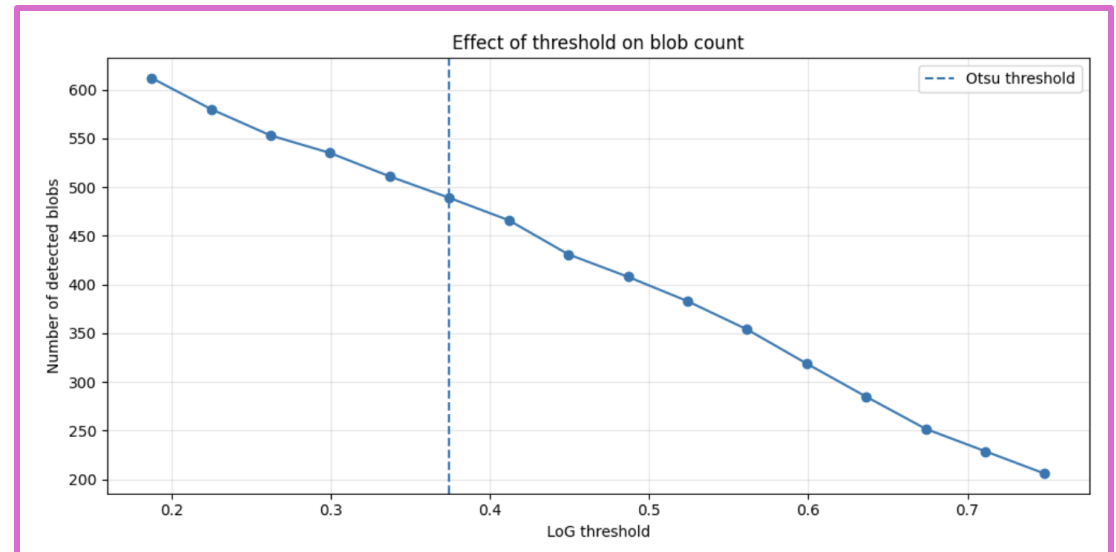
$$\sigma_B^2(T) = w_0 w_1 (\mu_0 - \mu_1)^2$$

proportion of object pixels

proportion of background pixels

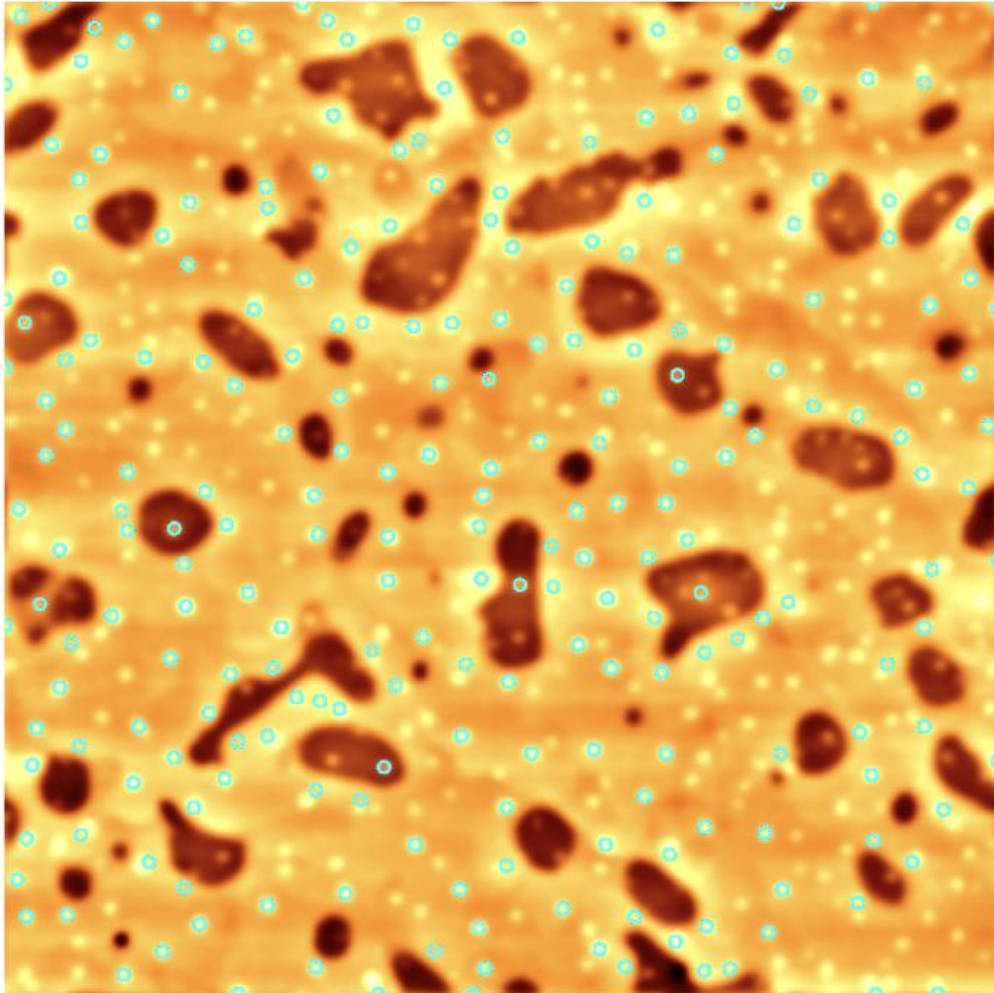
mean background intensity

mean object intensity

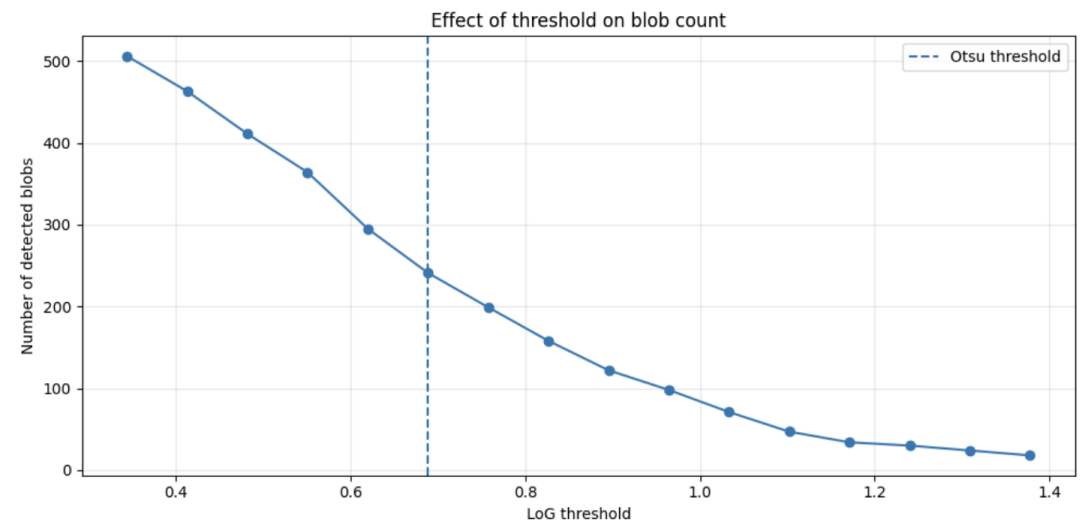
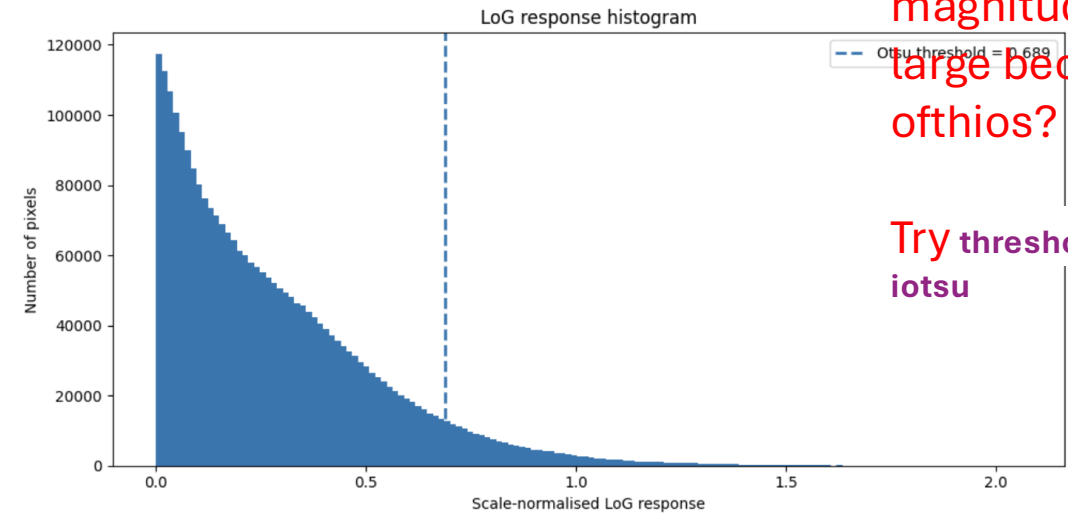


2.2 Thresholding – Otsu upper cutoff

Otsu threshold = 0.689
Detected blobs = 241



Restricted Otsu threshold: 0.6888598877711338



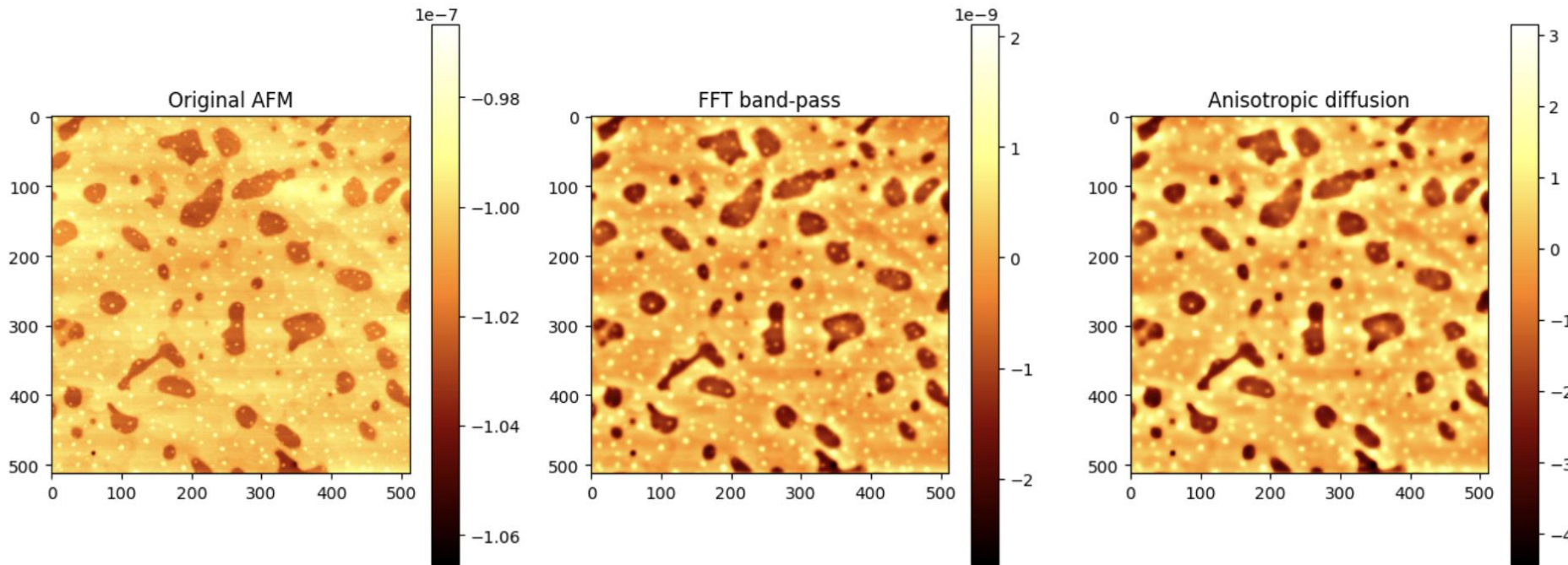
Otsu
causing backgr
ound issues ->
is Gabour
magnitude
large because
ofthios?

Try threshold_mult
iotsu

features that will be getting

3.1 Background removal techniques – anisotropic diffusion

Bandpass 20-150 pixels for the threshold, check that well in ImageJ) - INTERMO



Mean absolute change: 0.06522041901626291

Maximum absolute change: 0.5727873130697532

Correlation: 0.9961707063830949

$$\frac{\partial F}{\partial t} = \nabla \cdot (c(\nabla I) \nabla I)$$

The function c becomes small near strong edges $\therefore$ dot boundaries get preserved better.

$\rightarrow$ small local gradient $\therefore$ strong smoothing
 $\rightarrow$ large local gradient $\therefore$ weak smoothing

(unlike gaussian smoothing which is the same in all directions)

4.1 Other feature detection methods - Phase Congruency

Peaks before height filtering: 676

Candidate height range: 0.01692164121566393 0.28475894673132457

Phase candidates: 676

QDs at least 6 Å high: 497

Minimum accepted height: 6.022399999999915e-10

Median accepted height: 1.816128999999993e-09

Maximum accepted height: 3.304731999999996e-09

$$\therefore A_s = \sqrt{[e_s(x)]^2 + [o_s(x)]^2} = \text{local amplitude}$$

$$\varphi_s = \arctan 2(o_s(x), e_s(x)) = \text{local phase}$$

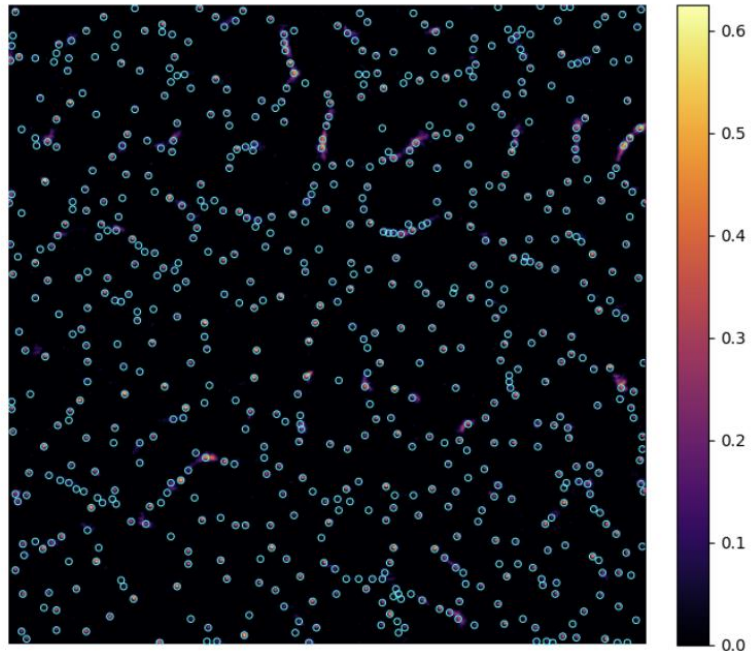
$$\therefore \text{local energy} = E(x) = \sqrt{\left[\sum_s e_s(x)\right]^2 + \left[\sum_s o_s(x)\right]^2}$$

$$\text{phase congruency at } x = PC(x) = \frac{E(x)}{\sum_s A_s(x) + \epsilon} = \begin{cases} \approx 1 \\ \approx 0 \end{cases}$$

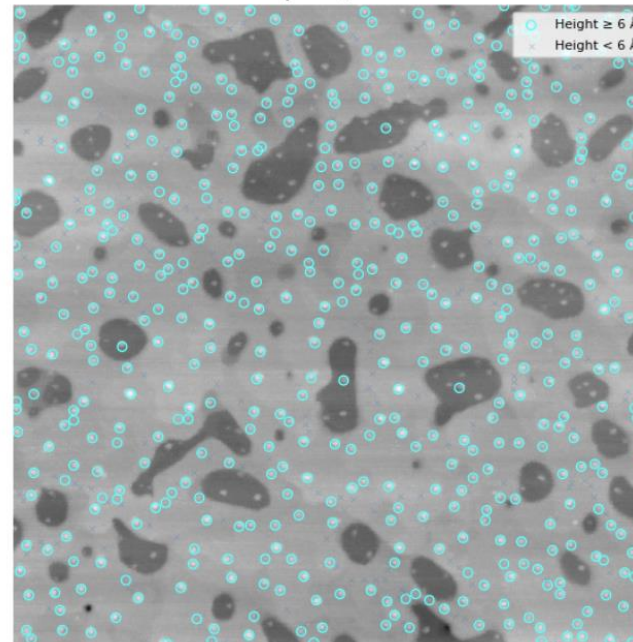
*all complex filters
responses point in
nearly same direction
→ congruent → edge*
*responses cancel
out: no
feature.*

*small constant to
avoid division by zero.*

Phase candidates: 676



Accepted QDs: 497



Try other radii for local background height

4.2 SIFT (Scale invariant feature transform)

The algorithm happens in 2 stages: first feature extraction, then description to select salient features.

This could help for looking out for other interesting effects or artefacts on an AFM scan.

SIFT examines many blurred versions of the image:

*$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$ where increasing σ removes more fine details.*

SIFT then subtracts nearby Gaussian-blurred images:

$D(x, y, z) = L(x, y, k\sigma) - L(x, y, \sigma) = \text{Difference of Gaussian} = \text{DoG}$ (which approximately behaves like a scale-normalised Laplacian of Gaussian),

$D(x, y, \sigma) \approx (k-1) \sigma^2 \nabla^2 L(x, y, \sigma)$

*$\therefore$ produces strong responses around blob-like structures whose size matches the current scale
.....*

If the DoG response is small, the candidate may be caused by noise:

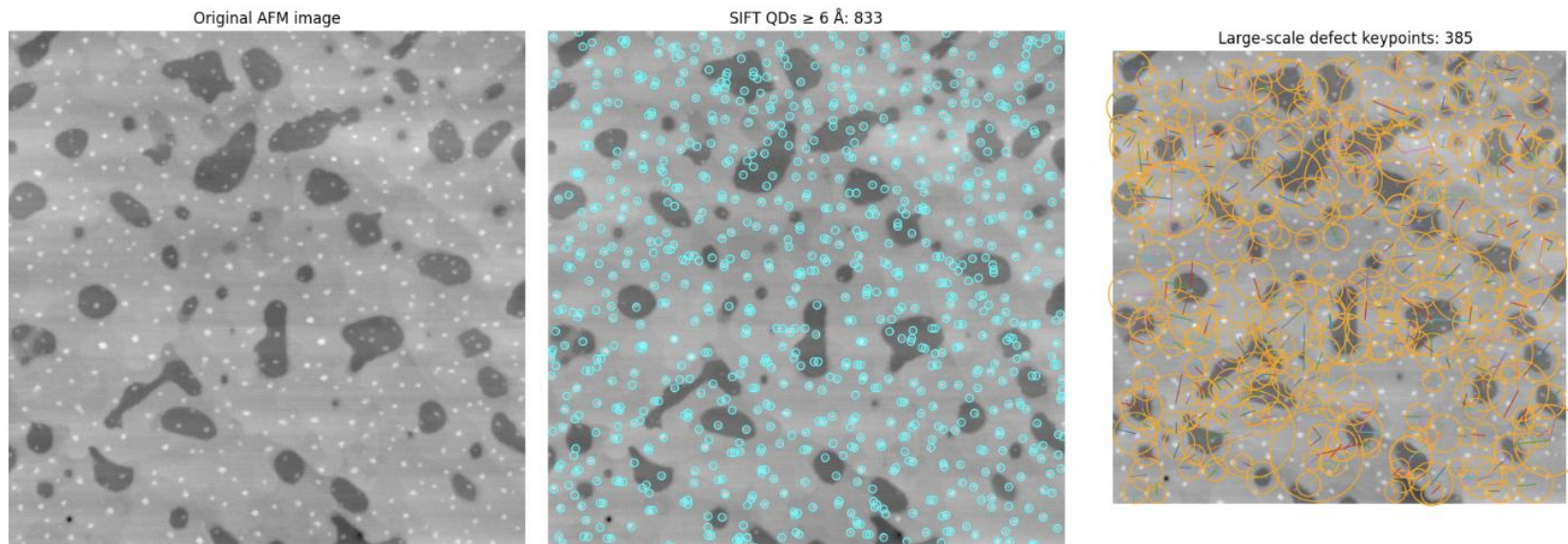
$|D(x, y, \sigma)| < T$

*Orientation = $\theta(x, y) = \text{atan2}\left(\frac{\partial L}{\partial y}, \frac{\partial L}{\partial x}\right)$
preserves direction of the slope unlike atan.*

- QDs: detect small bright structures after local-background subtraction, then require a physical height of at least 6 Å.
- Defects: detect larger dark structures using an inverted large-scale band-pass image.
- Other features?

4.3 Saliency for detecting other interesting features apart from the QDs

All QD-scale SIFT candidates: 4397
Accepted QDs at least 6 Å high: 833
QD descriptor array: (833, 128)
Large-scale defect SIFT keypoints: 385
Defect descriptor array: (385, 128)
QD height range: $6.0107749999999965 \times 10^{-10}$ to $3.3047319999999996 \times 10^{-09}$ m



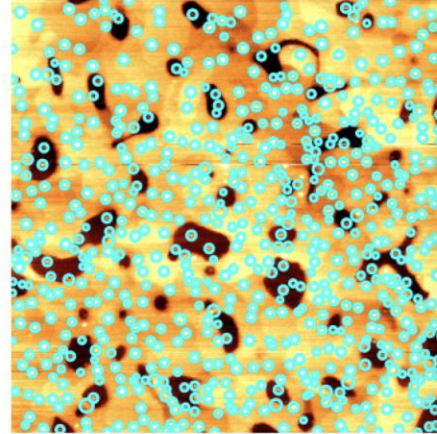
Gaussian smoothing and then Otsu for the defects, and better way to align the rows

Measure the height of the pits, and their depth

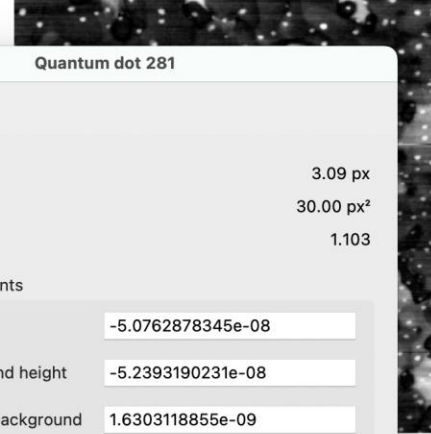
6.1 New QD app

Add the max of the
circle rather than of
the middle

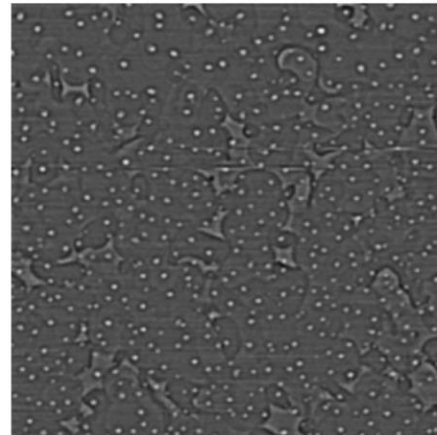
AFM image | features: 669



Top-hat



Top-hat + LoG response



Quantum dot 281

Quantum dot 281

Radius	3.09 px
Area	30.00 px ²
Circularity	1.103

Height measurements

Dot height	-5.0762878345e-08
Average background height	-5.2393190231e-08
Dot height above background	1.6303118855e-09

Notes / future data

Save Close

7.1 Training the segmentation model

- Code works, but I'm in process of labelling more dots now!
- Create ground truth to compare with LoG, GHT, Watershed and the segmentation model
- F1 score for all models
- Add a way to add dot detection selection bar into the QD app